

External manufacturing package

“Field Ambience Source files”

Package	Field Ambience Source files.zip — standalone KiCad 9.0.4 project “FIELD AMBIENCE”, gerbers, BOM.xlsx, CPL.xlsx, handoff PDF, 2 screenshots
Board	198.84 × 106.92 mm · 4 layers · 1.6 mm · 195 footprints · 239 nets · 166 vias
Audit date	2026-07-30 — revision 2 (revision 1 corrected, see §10)
Reference	AMBIENT repo state r19.42 · kicad/generate_kicad_project.py · firmware-c-next/src/ha1_h743/
Verdict	DO NOT FABRICATE — 6 blockers, 16 important findings, 7 regressions

Method

The netlist was reconstructed from the pad-to-net assignments inside `.kicad_pcb` (195 footprints, 239 nets, 1462 track segments, 166 vias) and symbol pinouts were extracted from `.kicad_sch` (302 symbols). Every pin assignment was then mirrored against two independent references that already exist in this project: the datasheet-verified symbol definitions in `kicad/generate_kicad_project.py` (TI SLUS810N, SLVSD38C, SLVSD76C; Diodes AP7361C and PAM8406) and the running firmware in `firmware-c-next/src/ha1_h743/`. On top of that: numeric geometry checks (zone-to-pad clearance measured across 583 pad instances, component spacing), gerber / drill / job-file inspection, and a designator cross-check between PCB, BOM and CPL.

This layout was NOT produced by `kicad/generate_kicad_project.py`.

It is a separate, hand-drawn project built on EasyEDA-imported libraries. That matters for reading the findings below: it also falls behind several decisions this project had already closed and verified (see §3, Regressions). `PCB_LAYOUT_STATUS.md` still claimed “no `.kicad_pcb` exists” (dated 2026-06-11) — that is now corrected in the repo.

1 · Blockers — do not fabricate

Six findings that either destroy hardware on first power-up, or leave a function permanently non-working. Two of them (B1, B2) sit in the same 5 V branch and should be fixed as one re-layout. A seventh, B5, appeared in the first revision of this report and was **withdrawn on re-check as a false finding** — see §10.

B1	L1 is a 1.8 μ H 0402 chip inductor in the boost power path
Finding	L1 = MLI1005F-1R8KT (LCSC C2885840), footprint L0402 — 0.54 × 0.54 mm pads — sitting on the switch node of U1 (TPS61089).
Why it matters	U1 boosts VBAT (3.0–4.2 V) to 5 V. At only 500 mA of 5 V load the peak inductor current already exceeds 2 A. A 0402 multilayer chip inductor is specified for a few hundred milliamps and saturates far below that; its DCR is in the hundreds of milliohms.
Failure mode	The converter never reaches regulation, sits in permanent current limit, and the inductor is thermally destroyed. The entire 5 V branch hangs off this — amplifier, headphone amp, and the 3.3 V LDO that feeds the MCU.
Fix	Use the part already in the repo BOM: SWPA6045S2R2NT , 2.2 μ H, 6045, ~4 A (C36500). While doing so, re-check the type-II compensation (R3/C6) — it was dimensioned for 2.2 μ H, not 1.8 μ H.
B2	The entire boost input current runs through slide switch SW2
Finding	VSY5 → SW2.1 / SW2.2 → PWR_ON, and PWR_ON feeds L1.1 and U1.9 (VIN) as well as U1.7 (EN) and U5.3 (ON).
Why it matters	SW2 is an SSSS811101 (C109335), an ALPS signal-level slide switch rated by LCSC at 12 V / 50 mA. 1.5–2 A flow through it here.
Failure mode	The contact welds shut or burns open — the unit ends up either permanently on or permanently dead. On top of that, a mechanically bouncing contact now sits directly in a switching converter's input path, which will restart the converter on every actuation.
Fix	Adopt the topology from ADR-0016 : VSY5 goes directly to U1.VIN and the input capacitor. SW2 switches only the PWR_ON signal (EN of U1 plus ON of U5).
B3	PWR_ON has no pull-down resistor
Finding	SW2 is SPDT and its second throw (pad 3) is unconnected. Net PWR_ON carries only C2, L1.1, U1.7 (EN), U1.9 (VIN) and U5.3 (ON) — no resistor to GND .
Why it matters	With the switch open, the boost EN and the load-switch ON inputs are left floating.
Failure mode	The device switches itself on from coupled noise or oscillates around the enable threshold; there is no defined microamp quiescent state when “off”.
Fix	R_PWR_PD (100 k) from PWR_ON to GND — same edit pass as B2.
B4	USB-C: D+/D– wired for one plug orientation only
Finding	USBC1.B6 = DR+, USBC1.B7 = DR–; pads A6 (DP1) and A7 (DN1) are unconnected — the PCB reports them as unconnected- (USBC1-DP1-PadA6) and -DN1-PadA7.
Why it matters	On a USB 2.0 receptacle A6 must be tied to B6 and A7 to B7, otherwise only one of the two plug orientations carries data.
Failure mode	USB — DFU flashing, charger detection, host communication — fails to enumerate in roughly half of all insertions. This is exactly the class of fault that costs days of bring-up time because it reads as a “flaky board”.
Fix	Connect A6 to B6 and A7 to B7 at the connector.

B5**PCA9685 drives LEDs against +5 V while firmware selects totem-pole****Finding**

U11 runs from 3.3 V (pin 28) and the LED anodes sit on the un-gated **+5 V** rail through 390 Ω .
 mcp23017_h743.c:197 writes PCA_MODE2 = PCA_MODE2_OUTDRV.

Why it matters

The PCA9685 LEDn outputs are 5.5 V tolerant **only in open-drain mode**. In totem-pole mode the absolute maximum is $V_{DD} + 0.5\text{ V} = 3.8\text{ V}$, and the 5 V pull-up continuously back-feeds the high-side driver and its body diode into the 3.3 V rail.

Failure mode

Latch-up risk on the PCA9685, LEDs glowing when nominally off, and injection into the 3.3 V supply.

Fix

Either set MODE2 = 0x00 (open-drain) in pca_dev_init(), which makes the existing wiring datasheet-compliant, **or** move the LED anode rail to 3.3 V and re-compute the series resistors. The firmware one-liner is the cheap path, but it has to land before bring-up.

B6**The two jack sockets contradict each other, and MIDI cannot drive****Finding**

CN2 and CN5 are the same part (PJ-320D, C431535) with the same footprint. On **CN2**, GND is on **pad 3**, audio L/R on pads 1/2, and the switch contact (jack detect) on pad 4. On **CN5**, GND sits on **pad 1**, R50 (10 k to 3.3 V) on pad 2, R49 (220 Ω from USART2_TX) on **pad 4**, and **pad 3 is unconnected**.

Why it matters

Two independent, provable points. (1) Contradiction: two instances of the same footprint place GND on different pads (3 vs 1). Whatever the true pad-to-contact mapping is, **both cannot be right** — that is pure netlist arithmetic and rests on no datasheet assumption. **(2) Arithmetic:** through 10 k Ω a 3.3 V leg can source at most 0.33 mA. A MIDI current loop needs 5 mA. R50 falls short by a factor of 15 regardless of which contact it lands on.

Not claimed

Which of the two jacks is the wrong one — that cannot be settled without the PJ-320D pad-to-contact mapping. The footprint geometry (pads 2/3/4 along the bottom edge, pad 1 alone on the top edge, barrel at -x) is not sufficient to decide it.

Failure mode

MIDI out dead. On top of that CN5 leaves a contact floating (pad 3), and on one of the two jacks a signal sits on the wrong contact.

Fix

Pin down the PJ-320D pad-to-contact mapping against the datasheet (flagged **UNVERIFIED — NEEDS HUMAN CHECK**), then rewire both jacks consistently. MIDI: tip $\leftarrow$ 33 Ω $\leftarrow$ USART2_TX, ring $\leftarrow$ 33 Ω $\leftarrow$ 3.3 V, sleeve = GND (repo reference r18.67 / ADR-0004).

2 · Important — resolve before fabrication

#	Finding	Impact	Fix
I1	STM32 VBAT (pin 6) is unconnected	ST requires VBAT tied to VDD when no backup cell is fitted. The backup domain and RTC are unreliable and start-up problems are possible — note that X2 (32.768 kHz LSE) is populated.	VBAT → 3.3 V + 100 nF
I2	VREF+ (pin 20) only decoupled by C43/C44, not tied to VDDA	The ADC has no reference; the pin floats unless firmware enables the internal VREFBUF.	VREF+ to the VDDA net, as the repo generator does
I3	Encoders 3 and 4 sit on pins with no timer quadrature — SW14 → PC4/PC5, SW15 → PB0/PB1	On the H743, PC4/PC5 have no TIM alternate function at all, and PB0/PB1 only offer CH3/CH4/CHxN — no TIMx_CH1/CH2 . The firmware runs all four encoders in TIM encoder mode (TIM2/3/4/1). Two of them would have to move to EXTI/polling; lost steps on fast rotation plus extra ISR load next to the audio hot path.	Move to PA0/PA1, PC6/PC7, PD12/PD13, PA8/PA9 (or the TIM2/TIM3 remaps PA15/PB3, PB4/PB5)
I4	Pin-map collisions with hal_h743 (detail in §4)	The board is not drop-in for the current firmware.	Plan a firmware port, or align the layout
I5	8 of 15 status LEDs are on a different PCA9685 channel than their net name implies	leds.c writes “LED n → channel n”, so the wrong LED lights.	Remap in firmware or fix the schematic (§5)
I6	Layer stack deviates from SPEC §9 (Sig/GND/PWR/Sig)	In1.Cu is a 3.3 V pour that is additionally sliced by 233 signal segments ; In2.Cu has no pour at all , only 151 tracks. The only ground is the F.Cu/B.Cu pours, which routing cuts apart. For a 480 MHz H7 plus a 2 MHz boost converter plus analogue audio, this is the single largest EMC risk in the layout.	Continuous GND plane on an inner layer; move signals off the plane layer
I7	DRC is effectively disabled in the project file	clearance, annular_width, hole_clearance, padstack, *_courtyard, silk_* and starved_thermal are all set to ignore , plus min_clearance = 0, min_track_width = 0, min_connection = 0. A DRC run against this project file proves nothing.	Restore severities to error and re-run DRC
I8	NPTH holes exported as plated holes with zero annular ring	The NPTH drill file is empty ; every mounting and anchor hole (CN2/CN5 1.0 mm, SW1/SW3–6 2.0 and 3.5 mm, SW2 0.9 mm, USBC1 0.75 mm) is in the PTH file with pad diameter equal to drill diameter.	Convert those pads to np_thru_hole in the footprints and re-export; otherwise expect a JLC DFM query
I9	Boost input capacitor 6.2 mm away from U1 (C2, 22 µF; output bank 4.6–9 mm)	The hot loop is far too large for a converter switching up to 2.4 MHz — overshoot on the switch node and EMC problems.	Re-place the power stage (already required by B1/B2)
I10	HSE crystal X1 on the opposite board side	~10 mm from PH0/PH1 and placed in the middle of the front panel. Electrically: a long, via-stitched clock path. Mechanically: an 11.4 × 4.7 mm HC49-SMD package sitting between the keys on the user-facing surface.	X1 plus C54/C55 directly at PH0/PH1 on the MCU side
I11	No series damping resistors on the LCD SPI	H1 is 54.5 mm from the MCU plus a cable to the module — ringing and EMI at the refresh rates the LCD-motion work targets.	22–33 Ω in series on SCK/MOSI/CS/DC near the MCU
I12	H1 pin order is non-standard (1 = VCC, 2 = GND, 3 = DIN, 4 = CLK, 5 = CS, 6 = DC, 7 = RST, 8 = BL)	Common ST7789 modules are GND, VCC, SCL, SDA, RES, DC, CS, BLK — plugging a stock module straight in puts VCC on GND. The backlight circuit (low-side 2N7002 on BL) also assumes BL is the LED cathode , whereas many modules expect BL to be an active-high enable.	Exact display module: UNVERIFIED — NEEDS HUMAN CHECK , then fix pinout and backlight path

#	Finding	Impact	Fix
I13	Battery NTC R13 is on the PCB, not on the cell	R13 (NCP15XH103) sits in the power corner next to the charger and the boost converter, so the BQ24074 TS input measures board temperature. Under load this can trip a temperature fault and abort charging. The repo deliberately uses a fixed 10 k resistor instead of an NTC.	Fixed 10 k, or a pack NTC at the cell via CN1
I14	PCA9685 EXTCLK (pin 25) floats	CMOS input; NXP specifies it must not be left open.	Tie to VSS
I15	PSRAM exposed pad (U9 pad 9) unconnected	Thermal and EMC.	Tie to GND
I16	Library part XL-1608U0C-06 (LED17) is self-contradictory — the symbol declares pin 1 = K, but footprint LED0603-R-RD_ORANGE puts pad 1 on the anode end (silk polarity chamfer at -x, where pad 2 sits)	The copper is correct — LED17 is wired the right way round. But the schematic shows “K” on the +5 V side, so KiCad net names and ERC misreport it, and LED placement orientation at the assembler is referenced to pad 1.	Straighten out the symbol/footprint pairing. Before ordering, confirm with the assembler whether LED rotation for C965800 references pad 1 or the chamfer

3 · Regressions against the repo state (r19.37 ff.)

Decisions this project had already made, verified and documented, which the external layout does not carry.

#	Board	Repo state	Consequence
R1	U8 = PAM8403DR (C17337)	PAM8406DR (C86270), r19.37 / ADR-0025 — the PAM8403 is NRND	Beyond the lifecycle question: the PAM8403 has a fixed +24 dB gain. The r19.37 gain-staging fix (R1 20 k → 174 k, +23 dB → +4.3 dB) cannot be applied to this board at all — the 5 V BTL amp clips in the analogue domain far below DAC full scale. This is precisely the bug that was already found and fixed.
R2	C69/C70 = 1 µF input coupling caps	10 nF (speaker HPF ~91 Hz, r19.37)	The high-pass now sits near 8 Hz; the 8 Ω 40 mm driver receives the full bass energy.
R3	TPA6132A2 gain set to 0 dB (G0 = H, G1 = L)	−6 dB per ADR-0024	Re-assess the level plan — together with R1.
R4	U3 = AP7361C-33E-13, SOT-223 (C500795), wired as 1 = IN, 2 = GND, 3 = OUT, tab = GND	AP7361C-33Y5-13, SOT-89-5 (C460397); r18.6 explicitly rejected a SOT-223 variant because its pinout differs	AP7361C SOT-223 pin order: UNVERIFIED — NEEDS HUMAN CHECK. If wrong, the entire 3.3 V rail is dead. Must be checked against the Diodes datasheet before fabrication.
R5	U9 = APS6404L-3SQR-ZR, USON-8 (C3040877)	APS6404L-3SQN-SN, SOIC-8 (C3028887)	Pin mapping checked and correct ; only the BOM divergence needs documenting.
R6	U2.14 (TMR) → GND = safety timer disabled	TMR = NC = 5 h default	Intentional? Otherwise leave NC.
R7	—	PCB_LAYOUT_STATUS.md: “no .kicad_pcb exists” (2026-06-11)	Repo documentation was stale and has been corrected.

4 • Firmware ↔ board pin map

Compared against `firmware-c-next/src/hal_h743/`.

Matches

Function	Firmware	Board
SAI1 FS / SCK / SD	PE4 / PE5 / PE6	pin 3 / 4 / 5
QSPI CLK / BK2_NCS / IO0-3	PB2 / PC11 / PE7-PE10	pin 36 / 79 / 37-40
Amp /SHDN, /MUTE	PB14, PB15	pin 53, 54
MIDI TX	PD5 (USART2)	pin 86
LCD SCK / MOSI / CS	PA5 / PA7 / PA6	pin 29 / 31 / 30
Encoder 1, 2	PA0/PA1 (TIM2), PC6/PC7 (TIM3)	match — A/B swapped on enc 2, so rotation is inverted
MCP23017 GPA0-4 = cells, GPA5 = XSMT, GPA6 = jack, GPA7 = VBUS	IODIRA 0xDF, GPPUA 0x5F	bit-exact match
Backlight = PCA9685 channel 15 → 2N7002	yes	yes

Diverges

Function	Firmware	Board	Note
LCD DC / RES	PC4 / PC5	PD15 / PD14	PC4/PC5 are now encoder 3 on this board
MCP23017 bus	I ² C1 PB6/PB7 (shared with PCA9685)	I²C4 PD12/PD13	PD12/PD13 are TIM4 = encoder 3 in firmware
Encoder 3	PD12/PD13 (TIM4)	PC4/PC5	see I3 — no timer quadrature
Encoder 4	PA8/PA9 (TIM1)	PB0/PB1	see I3 — no timer quadrature
Encoder push 1-3	PE0 / PE1 / PE3	MCP GPB5-7 + PD4	port required

Conclusion: no respin needed for the pin map alone — but a planned firmware port is, with the single exception of I3, which is a hardware limitation.

5 • LED ↔ PCA9685 channel (as built)

Net name	Actual channel		LED
LD0-LD4	0-4	✓	LED6, LED4, LED5, LED3, LED7
LD5	11	✗	LED16
LD6	6	✓	LED9
LD7	10	✗	LED10
LD8	8	✓	LED11
LD9	7	✗	LED13
LD10	5	✗	LED8
LD11	14	✗	LED14
LD12	13	✗	LED15
LD13	12	✗	LED12
LD14	9	✗	LED17 (also reversed — see B5)
LD15	15	✓	backlight FET Q1

6 • Cosmetic and documentation debt

Violations of `AI_READY_SCHEMATIC_STANDARD.md` — not fabrication-critical, but they make the schematic unusable as machine-readable design data.

- **Net names containing spaces:** LCD CS, LCD DC, LCD SCK, LCD RST, SCL MCP, SDA MCP, SCL PCA, SDA PCA, OUT DR+, OUT DR-. Active-low suffixes are missing throughout: SHDN → AMP_SHDN_N, MUTE → AMP_MUTE_N.
- **Net labels contradict the actual pins:** “PB0”/“PB1” on pins 35/34 (swapped), “PC6”/“PC7” on pins 63/64 (swapped), “GPA5” on MCP pin 25 (which is GPA4).
- **Wrong MPN / package fields in the schematic.** The BOM is correct in every one of these cases, so nothing gets mis-ordered — but the schematic lies: R10/R14 read “470 Ω ” with MPN field RC0603FR-07470KL = **470 k Ω** (BOM: C144657 = 470 Ω ✓); R8/R53 read “200 k Ω ” with MPN field 0603WAF3900T5E = **390 Ω** (BOM: C105574 = 200 k ✓); R37/R38/R55/R56 carry footprint field R0402 with an 0603 MPN (BOM: C99782 = 4.7 k 0402 ✓, and the pads really are 0402 ✓); the C39 family uses a library named “10uF 0805” on a C1210 land.
- **Symbol typos:** PAM8403 pin 12 is named “SHND”; the MCP23017 symbol names pin 12 “SCK” (the SPI name of the MCP23S17 — here it is SCL).
- **The BOM lists H2-H5 (mounting holes) as line items with an empty LCSC field.** The JLC BOM upload will error on those. Delete them or mark DNP.
- copper_finish: “None” in the stack-up, and correspondingly “Finish”: “None” in the gerber job file. Set ENIG/HASL so the fabrication data matches the order.
- **Unsaved schematic edits inside the archive:** _autosave-FIELD AMBIENCE.kicad_sch (30 Jul, 14:23) is **newer** than FIELD AMBIENCE.kicad_sch (29 Jul, 22:39) and additionally contains the Mechanical:MountingHole_Pad symbols (H2-H5). The saved schematic is therefore behind the board by exactly those four holes. The .kicad_pcb is identical to its autosave ✓.

7 · How solid is each finding?

Not every item above rests on the same footing. Anyone re-checking this report should know where to start.

Class A — read straight out of the delivered files.

No datasheet knowledge, no judgement. Open the files and you see the same thing: B2, B3, B4, I1, I5, I6, I7, I8, I9, I10, I16, the contradiction in B6, everything in §6, and everything in §8. For B4 the board even spells the missing connection out: unconnected- (USBC1-DP1-PadA6).

Class B — file plus simple arithmetic.

B1 (package from the BOM and the pad dimensions — that a 0402 chip inductor will not carry 2 A is not a borderline call), the 10 kΩ calculation in B6, I9.

Class C — depends on datasheet figures recalled from memory. Please verify these.

#	The assumption	Why it still holds / how to check
B2	The SSSS811101 is rated around 12 V / 50 mA	The exact number needs checking. Qualitatively it changes nothing: an ALPS signal-level slide switch is not a 2 A power switch.
B5	PCA9685 LEDn pins are 5.5 V tolerant only in open-drain mode; in totem-pole the limit is VDD + 0.5 V	The one I would most want independently checked. See NXP PCA9685, “Limiting values” together with MODE2 / OUTDRV.
I3	PC4/PC5 have no TIM alternate function on the H743; PB0/PB1 only CH3/CH4/CHxN	Check against DS12110 “Alternate function mapping”. If a CH1/CH2 does exist there, half the finding falls away — the collision with hal_h743 from I4 remains either way.
R1	The PAM8403 has a fixed +24 dB with no external gain resistors	Supported by this project’s own history: r19.37 / ADR-0025 moved to the PAM8406 with an external RI for exactly this reason.

Two cross-checks that give the MCU analysis a second witness

The STM32 symbol in the delivered project (STM32H743VIT6_C114409) was compared pin by pin against the pin table this repo had already verified from DS12110: **60 of 60 checked pins agree**, the only difference being the names “VCAP” versus “VCAP1”/“VCAP2”. I1 (VBAT open) and the whole pin-map comparison in §4 therefore rest on two independent sources.

Pad numbering verified, so no finding rests on a silent footprint assumption

After the revision-1 error (§10), every finding that could depend on how a footprint numbers its pads was re-checked against the geometry: the **LQFP-100** carries 100 pads, 4 × 25 contiguous per side; the **TSSOP-28** is numbered in the standard wrap-around order. That secures I1 and I5. **SW2**’s pad 2 is geometrically the centre terminal, i.e. the changeover pole, so VSYS really does reach PWR_ON through the switch — which secures B2 and B3. And the polarity of **all 17 LEDs** was recomputed at footprint level: all 17 are the right way round on the board.

Three findings that were wrong, or would have been

Recorded for honesty — the same review process that produced the list above also produced these.

- **R_FSW returning to the SW node** looked like a wiring error. It is **correct** (TI SLVSD38C table 6-1) — checked against the repo generator and dropped before it ever reached the report.
- **ITERM (U2 pin 15) left open** looked like a missing resistor. It is the **datasheet default** (10 % termination) — likewise dropped beforehand.
- **“LED17 reversed”** was a **genuine false finding** and stood as a blocker in revision 1 of this report. It came from reading the symbol pin names without looking at the polarity marking in the footprint. See §10.

8 · What checked out clean

Verified, no objection — recorded so the report is not read as “everything is broken”.

- **BOM ↔ CPL ↔ PCB are 1:1 consistent:** 195 designators in board and BOM, no orphan in either direction, every line quantity equals its designator count, LCSC numbers populated throughout (except H2-H5), CPL correctly holds 191 rows (holes excluded), and top/bottom assignment agrees between CPL and PCB (69 top / 122 bottom).
- **No floating nets:** all 239 nets have at least two pads.
- **No pour shorts:** despite connect_pads (clearance 0) on the GND zone, the measured fill polygon keeps **exactly 0.2 mm** to every foreign pad — 583 pad instances checked, median 0.201 mm.
- **STM32 decoupling is good:** one 100 nF within 1.8–3.3 mm of each of the five VDD pins; VCAP1/VCAP2 with 2.2 µF at 2.8 / 3.2 mm.
- **BQ24074 correctly configured** (against this project's verified reading of TI SLUS810N): EN1 = LO / EN2 = HI selects resistor-programmed ILIM ✓; CE_N = GND ✓; **ITERM = NC is the datasheet default (10 %) and therefore correct** ✓; ISET 1.13 k ≈ 0.79 A; ILIM 1.18 k ≈ 1.36 A; EP on GND ✓; /CHG and /PGOOD LEDs correctly polarised ✓.
- **TPS61089:** FB divider 332 k / 107 k → 4.94 V ✓; **R_FSW returning to the SW node is correct** per TI SLVSD38C table 6-1 — it looks wrong but is not ✓; BOOT capacitor present ✓.
- **TPS22918:** QOD tied to VOUT, CT left open — exactly the ADR-0016 intent ✓.
- **TPA6132A2:** charge-pump network complete (CPP/CPN/HPVDD/HPVSS), single-ended inputs with IN+ to GND, EP on GND ✓.
- **PCM5102A / APS6404L / MCP23017 / PCA9685:** pin assignments checked against the symbols, no transposition.
- **LED polarity of all 17 LEDs verified at footprint level** (silk chamfer = cathode, cross-checked across all four LED footprints): **all 17 are the right way round on the board** ✓. Only LED17's symbol/footprint pairing is inconsistent (I16).
- **USB ESD (USBL6-2SC6) present, CC1/CC2 with 5.1 k** ✓. D+/D− pass through D1 channel-clean (pin 3→4 and pin 1→6) ✓.
- **Board outline closed**, 198.84 × 106.92 mm, 4 layers, 1.6 mm, vias 0.6/0.3 mm, inner-layer clearance 0.127 mm — all within standard JLC capability.

9 • Recommended order of work

	Items	Action
1	B1 + B2 + B3 together	Redraw and re-place the power stage: inductor, switch path, pull-down, input capacitor at the IC.
2	B4, B6	Pure netlist corrections.
3	B5	Firmware one-liner (MODE2 = 0x00) — can be done now.
4	R1 + R2 + R4	Bring the audio front end up to the r19.37 state and verify the LDO pinout.
5	I1, I2, I14, I15, I16	Small netlist and library corrections.
6	I3 + I4	Straighten out the encoder pins, then plan the firmware port.
7	I6	Layer stack to SPEC §9.
8	I7 + I8	Restore DRC severities, fix NPTH, re-export gerbers — only then is a DRC run meaningful.

Open UNVERIFIED — NEEDS HUMAN CHECK items, per the project's schematic standard: the PJ-320D pad-to-contact mapping (B6), the AP7361C SOT-223 pin order (R4), the exact display module behind H1 (I12), and the SSSS811101 land pattern (§2, addendum).

10 · Changes from revision 1

Revision 1 of this report listed **seven** blockers. On challenge, it was re-checked.

“LED17 reversed” was an audit error, not a layout error — withdrawn.

It was derived from the symbol pin names alone, without checking the polarity marking in the footprint. Recomputed across all four LED footprints, the silk chamfer consistently marks the cathode — and on LED17 that end carries pad 2. So the schematic wiring (pad 1 through R45 to +5 V, pad 2 to the PCA sink) puts the anode on +5 V: **the copper is correct and LED17 lights.**

Part	Symbol pin 1	Pad 1 at	Silk chamfer (cathode)	Symbol vs footprint
KT-0603G	K	−x	−x → pad 1	consistent
KT-0603Y	A	+x	−x → pad 2	consistent
XL-1608UWC-04	K	−x	−x → pad 1	consistent
XL-1608U0C-06	K	+x	−x → pad 2	MISMATCH

The “-R-” footprint is the mirrored variant. Three of the four parts pair it consistently; XL-1608U0C-06 does not. What remains is that library inconsistency, recorded as **I16** — important, not a blocker.

- **B6 (MIDI / jack sockets) was softened.** The claim that CN5 sits on the switch contact was an inference from assuming CN2 was the correct one. The finding now rests only on what is provable without the PJ-320D datasheet: the two sockets contradict each other on which pad is GND, and 10 kΩ cannot source a 5 mA loop from 3.3 V.
- **Blockers renumbered.** Old B6 → B5, old B7 → B6. No gap in the sequence.
- **New §7, “How solid is each finding?”** Every item is now classified by what it rests on, including the four that depend on datasheet figures recalled from memory and should be independently checked.
- **New cross-checks in revision 2.** Pad numbering of the LQFP-100 and TSSOP-28 verified (securing I1 and I5 against silent footprint assumptions); SW2's pole assignment confirmed geometrically (securing B2 and B3). In the process the irregular SW2 pad pitch surfaced as a new item.

Two further suspicions were discarded **before** revision 1 and never reached the report: R_FSW returning to the SW node (looks like a wiring error, is correct per TI) and ITERM left open (the datasheet default). Both are now listed under §8 “clean”, so they do not resurface as findings on the next review.